STATEMENT OF WORK (SOW)

1.0 TITLE

Cancer Research Data Commons (CRDC) Data Discovery Portal

2.0 BACKGROUND

General Overview
One of the overarching goals of the Cancer Moonshot initiative is to enhance data sharing. One of the key recommendations from the Cancer Moonshot's Blue Ribbon Panel (BRP) is to build a National Cancer Data Ecosystem. The objective of this Ecosystem is to:
"Enable all participants across the cancer research and care continuum to contribute, access, combine, and analyze diverse data that will enable new discoveries and lead to lowering the burden of cancer."

The National Cancer Data Ecosystem is envisioned to include:

- Enhanced cloud-computing platforms
- Services that link disparate information, including clinical, image, and molecular data
- Essential underlying data science infrastructure, methods, and portals
- Sustainable data governance to ensure long-term health of the Ecosystem
- Standards and tools to enable interoperable data

NCI's primary contribution to the National Cancer Data Ecosystem is the CRDC, a virtual, expandable infrastructure that provides secure access to diverse data types across scientific domains and the ability to perform cross-domain analysis of large data sets that can ultimately lead to new discoveries in cancer prevention, treatment, and diagnosis. The CRDC co-locates data and computing infrastructure with commonly used services, tools, and applications for analyzing and sharing data to create an interoperable resource for the cancer research community. The CRDC consists of multiple infrastructure components focused on data interoperability and analysis, as well as multiple data repositories, each currently with its own web portals, Application Programming Interfaces (APIs), and submission systems and guidelines.

Enabling easy access to the breadth of data by as many scientists as possible helps the cancer research community to make new discoveries, learn from every cancer patient, and improve cancer outcomes. One of our most critical challenges is the lack of easy-to-use data exploration and visualization tools for scientists of all backgrounds, including those who have no computational training, within the CRDC ecosystem.

NCI has partnered with the Advanced Research Projects Agency for Health (ARPA-H) to build the ARPA-H Biomedical Data Fabric (BDF) Toolbox program to bring Artificial Intelligence (AI)-powered tools to the CRDC. The ARPA-H is a US government funding agency within the Department of Health and Human Services (HHS) that supports high-impact research capable of driving biomedical and health breakthroughs that can deliver transformative, sustainable, and equitable health solutions for everyone. ARPA-H's mission focuses on leveraging research advances for real world impact. In collaboration with NCI CRDC, ARPA-H BDF program is developing prototype tools using cancer data as the first use case. Primary objective of the BDF program is to implement revolutionary approaches to help researchers collect data in a standardized, harmonized format to lower the barriers associated with data collection, reduce the time needed to integrate new data sources, and improve data usability across disciplines and biomedical literacy levels.

Currently, users access data hosted across the CRDC ecosystem through separate data repository-specific web portals. One of the most critical challenges is the lack of an intuitive cancer data search capability across the entire ecosystem to easily explore what multi-modality data is available for analysis. Enabling easy access to the breadth of data by as many researchers with varying degree of biomedical and technical literacy as possible helps the cancer research community to make new discoveries, learn from every cancer patient, and improve cancer outcomes. Therefore, NCI seeks to develop a unified CRDC Data Discovery Portal that will enable users to search, access, and visualize all CRDC data in a streamlined manner, greatly simplifying the user experience and lowering barriers to reusing data to advance cancer research.

The NCI seeks to recommend sole-sourcing the work to the ARPA-H BDF performer team from Jataware Corp. to integrate and expand their prototype BeakerHub platform into the CRDC ecosystem to create an AI-powered next-generation Data Discovery Portal that combines traditional faceted search with a conversational AI assistant powered by an integrated Jupyter notebook environment, enabling rapid data analysis in one setting for users.

CRDC Data Repositories

The CRDC currently has seven data repositories in production. The data repositories, except for the General Commons and the Integrated Canine Data Commons, each hold primarily datatype-specific datasets and contain open access, and/or controlled access data. Open access data include de-identified patient clinical and phenotypic data, biospecimen data, verified somatic variants, de-identified imaging data, mass-spec data, and other derived or summary data. Open access data are available to all with no restrictions. Controlled access data contain potentially identifiable information and are restricted to authorized users of the data and include, but are not limited to, any genomic data with a patient's germline information (e.g., BAM and FASTQ files, germline variant calls, non-validated somatic mutations). Users who wish to view controlled access data must apply with a scientific use case and be approved for use by an NIH committee. Within the CRDC repositories users are authorized by NIH's central Researcher Auth Service (RAS) via the Data Commons Framework (see below) to gain access to controlled data assets.

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- The Genomic Data Commons (GDC) was the first CRDC data repository to be implemented and serves as a repository for cancer genomic data from The Cancer Genome Atlas (TCGA), Therapeutically Applicable Research to Generate Effective Treatment (TARGET), Foundation Medicine Inc. (FMI), Clinical Proteomics Tumor Analysis Consortium (CPTAC), and many other important cancer projects. The GDC accepts genomic data including BAM and FASTQ files, as well as standardized clinical and biospecimen data, and contains both open and controlled access data. Genomic data in the GDC are harmonized across projects, as such enabling cross projects comparison, through running common pipelines, including the re-alignment of sequencing data against a common human genome build and the generation of standardized variant calls. The GDC provides search, visualization, and limited analysis capabilities for its users through the GDC portal (https://portal.gdc.cancer.gov/). GDC data are stored on-prem as well as in both the Google and Amazon clouds and may be accessed through the GDC graphical user interface portal, application programming interfaces (APIs), and the NCI Cloud Resource (see below).

- The Proteomic Data Commons (PDC) collects, harmonizes, and hosts proteomic data across a variety of tumor types. The PDC incorporates data from programs such as Clinical Proteomic Tumor Analysis Consortium (CPTAC) and International Cancer Proteogenome Consortium (ICPC) where the majority of data are generated from quantitative mass spectrometry-based proteomic targeted assays. The PDC data portal (https://proteomic.datacommons.cancer.gov/pdc/) provides a space to query, visualize, and download available datasets, all of which are open access. The PDC data are stored on Amazon Web Services and the data are also available through the NCI Cloud Resource.

- The Imaging Data Commons (IDC) makes available medical imaging data from multiple imaging modalities (e.g., CT, MRI, PET) in the cloud and includes data from The Cancer Imaging Archive (TCIA). IDC also hosts additional imaging datasets including digital pathology and cellular and molecular imaging. The IDC uses the DICOM standard as the primary file format for storage and the IDC data portal (https://portal.imaging.datacommons.cancer.gov/) provides searching, browsing, and filtering capabilities. The IDC currently holds only open access datasets, and the data are stored in the Google cloud. Imaging data in the IDC are also available through the NCI Cloud Resource.

- The General Commons (GC) is a repository for storing NCI-funded data that are currently not hosted by the other CRDC data repositories. The GC does not limit the type of data submitted and will take datasets including genomic, proteomic, and imaging data not currently managed by the GDC, PDC, and IDC, respectively, for reasons related to size, access or other constraints. Both open access and controlled access datasets are accepted by the GC. Data models can be searched within the GC data portal (https://general.datacommons.cancer.gov/#/) with participant and sample data available through the Genotypes and Phenotypes (dbGaP) database and sequence data available through NCBI SRA and the NCI Cloud Resource.

- The Integrated Canine Data Commons (ICDC) is a repository for a variety of data relating to cancer in dogs. All data in the ICDC are open access and contain a variety of data types and formats, such as genomic, imaging, clinical trials, and treatment outcomes data. The ICDC has several genomic datasets from canine cancer research studies available for query through the ICDC data portal (https://caninecommons.cancer.gov/) and plans to add more data from canine cancer research studies and canine clinical trials. The ICDC data are stored on Amazon Web Services and the data are also available through the NCI Cloud Resource.

- The Clinical Translational Data Commons (CTDC) was launched in 2024. The CTDC stores and shares clinical, biospecimen and molecular characterization data from cancer clinical studies. Currently the CTDC incorporates data from the Cancer Moonshot Biobank, with additional datasets to be released as they become available. The CTDC accepts both open and controlled access data, searchable through the CTDC data portal (https://clinical.datacommons.cancer.gov/#/). CTDC data are stored on Amazon Web Services and available through the NCI Cloud Resource.

- The Population Science Data Commons (PSDC) is the newest CRDC data commons, launching in early 2026. The PSDC has been designed to host several types of data, including but not limited to survey and questionnaire data, biomarker assays, environmental exposure measurements, dietary and anthropometric assessments and biometrics from exercise-capturing technologies. The PSDC hosts data from the NCI's Cancer Epidemiology Cohorts (CEC) as well as other NCI-funded research programs and awards. The PSDC accepts both open and controlled access data. Currently, the PSDC portal (https://populationsciences.datacommons.cancer.gov/#/ - NOT YET ACTIVE) allows for search at the comprehensive study level, with plans to implement participant-level data and search in a future release. PSDC data are stored on Amazon Web Services and available through the NCI Cloud Resource.

CRDC Infrastructure and Analysis Components

- The Data Commons Framework (DCF) provides central Authentication and Authorization (Fence) and digital ID (IndexD) services for the CRDC. The DCF also maintains the copies of the GDC genomic data in the Amazon cloud platform for access through the NCI Cloud Resource.

- The Cancer Data Aggregator (CDA) is a cloud-based infrastructure that provides a query engine via an Application Programming Interface (API) layer allowing researchers to aggregate diverse data types distributed across the CRDC. Using the

CDA users can discover, query, retrieve, and aggregate data according to a variety of search parameters. In addition to a query engine, the CDA provides a central search database for cross-cutting data that serves as a primary source of truth for the CRDC. It contains demographic, clinical and other descriptive data, henceforth to be referred to as metadata, in a structured format to support federation across multiple repositories. The CDA API with accompanying Jupyter notebooks is currently available to search and aggregate data across five CRDC repositories (GDC, PDC, IDC, ICDC and GC). CDA provides publicly accessible metadata only and therefore is a FISMA low resource.

- The NCI Cloud Resource (Seven Bridges Cancer Genomics Cloud [SB-CGC]) was designed to meet the cancer research community's need to analyze large-scale cancer data by eliminating both the need for researchers to store petabytes of data and the prohibitive cost and time required for download. The Cloud Resource harnesses the genomic, proteomic, and imaging datasets as well as their associated clinical and biospecimen data available throughout the CRDC in the Google Cloud Platform and Amazon Web Services to co-locate these data with the elastic compute power of these commercial clouds. The NCI Cloud Resource provides analytic tools, pipelines, and user workspaces for sharing data and collaborative analyses, and each allows researchers to bring their own tools and data to the cloud. The Cloud Resource utilizes the DCF to give users access to both open and controlled access data across the CRDC.

Rationale

The mission of the CRDC is to provide secure access to cancer research data and analytical tools by connecting multiple cloud-based data repositories and to serve as a central location to support public data sharing for NCI-funded programs. CRDC domain- and program-specific components (e.g., GDC, PDC, ICDC, IDC, CTDC, PSDC, GC) act as data repositories and web portals for search and data viewing or download while the NCI Cloud Resource provides workspaces for cancer researchers to conduct complex data analysis and computation in the cloud. The DCF provides the centralized Authentication and Authorization services in coordination with NIH RAS initiative to support safe and secure data access across the entire ecosystem. Cancer Moonshot programs such as Human Tumor Atlas Network (HTAN) and NCI-funded initiatives such as Childhood Cancer Data Initiative (CCDI) are actively depositing multi-modal data into the CRDC repositories for public data sharing where these data are made available through CRDC repository portals and the NCI Cloud Resource for analysis by the scientific community.

Currently, users access CRDC data through individual data commons-specific web portals. A critical challenge is the lack of user-friendly exploration across all data components of the CRDC, thereby limiting the scope of research-based questions that can be asked. Users lack the ability to effectively browse or search across all available data types across the CRDC, or to find subjects or samples with data from multiple modalities. The largest blocker to creating a multi-modal, cross-data commons search is the challenges associated with harmonization of clinical and phenotypic metadata stored in disparate data models across the individual commons that represent data using its own DC-specific standards.

The Cancer Data Aggregator, released in April 2024, attempts to address this challenge through a query engine on top of a central search database for cross-cutting data, enabling a minimum level of cross-query functionality across CRDC data commons. The most recent August 2025 software update introduced a new, streamlined, object-based query API and cdapython tool supporting a more intuitive query language, subject and file-based results tables, and support for joining data across multiple results, enabling more complex filter sets. Data is extracted via data commons specific APIs, transformed and aggregated, followed by a combination of automated and manual harmonization steps depending on the element in question. For some elements, ontologies are “slimmed” or values are manually rolled up to higher terms in the ontology to allow for more intuitive searching. Search is available through a minimal GUI, through cloud-hosted python notebooks, or through a local cdapython installation.

While a step in the right direction, several critical limitations continue to impede the ability of users to search and discover comprehensive cancer data. The limitations include:

- lack of an intuitive and dynamic user interface
- time required to learn the cdapython query language and to develop queries of interest
- assumption or expectation of previous notebook experience to run and save queries, which is a barrier to reproduction for a wide group of users
- limited sets of harmonized data elements due to the time required for manual harmonization approaches
- limited ontological slims or rollups, again due to manual lift required

The Data Discovery Portal aims to address the first three of these limitations, enabling users to search, access, and visualize all CRDC data in a streamlined manner, greatly simplifying the user experience and reducing barriers to data access and reuse. While the last two are outside of the scope of this work, we anticipate that parallel efforts on solutions for data harmonization and ontology mapping will feed into this work as part of the broader CRDC infrastructure.

Advancements in AI and new language models present an unprecedented opportunity to address the challenges of searching across multiple data types across different data commons with varying levels of harmonization. The success of implementing AI to create a comprehensive and powerful discovery portal requires that the underlying technology accurately extracts, represents, and reason over data across data repositories, enables search, and allows humans to intuitively interact with results. CRDC users have a vast range of understanding of data availability, as well as technical experience and knowledge. Given the wide range of user knowledge and experience, simply adding an AI-powered natural language chat on top of data repositories will not provide an inclusive solution to the above-stated challenges. The successful implementation of the Data Discovery Portal should provide users with options to explore CRDC data using traditional search methods alongside the AI-powered interactive agent, as agents are most useful when the user has an understanding of the data and resources available.

In the past two years, NCI has partnered with ARPA-H on a Biomedical Data Fabric Toolbox (BDF) program to develop innovative technologies to enhance data usability by the biomedical research community. Jataware, an awardee of the BDF program, has built a prototype platform, BeakerHub, that provides an AI-assisted notebook environment and has demonstrated transformative solutions for how researchers can explore and analyze data across multiple data repositories. Beakerhub is an large language model (LLM)-based, agentic platform for scientific discovery, with which users can interact via chat and code, generate Jupyter notebooks, readily use integrated domain-relevant knowledgebases, tools, and datasets, and leverage pre-built workflows. It allows users to search for data and perform tasks from within the same environment, leading to enhanced capabilities for analysis of multimodal data. The BeakerHub platform and LLMs were specifically designed for biomedical research and data analysis. Moreover, as part of the ARPA-H BDF program, Beakerhub has integrated with another BDF performer's tool from Northeastern University, the Integrated Network and Dynamical Reasoning Assembler (INDRA). INDRA collects and assembles extensive knowledge from publications and repositories, creating a computable graph that can be queried for a variety of biomedical information, enhancing Beakerhub's unique ability to develop a comprehensive search analysis platform. As CRDC is designated to be a transition platform for successful ARPA-H performers, BeakerHub has successfully tested integration of multiple CRDC Data Commons and CDA into its system. Beakerhub's architecture was developed to enable the rapid addition of new data sources, allowing for the implementation of a flexible and scalable Data Discovery Portal that will grow seamlessly as the number of data repositories and data types within the NCI data ecosystem continues to expand rapidly. BeakerHub connects directly to data sources APIs and has specialized agents that leverage existing tools such as a Data Repository Service (DRS) Uniform Resource Locator (URL) resolver to securely access cancer research data files directly allowing for analysis capabilities in one workspace. In addition to direct data connections, BeakerHub processes resource documentation to aid in providing the user with the necessary information to fully use data resources appropriately to further their analysis.

BeakerHub is unique as it will allow the CRDC an opportunity to provide the cancer research community with cutting-edge AI technology by providing a portal that significantly enhances the common faceted search. The CRDC considers faceted search a requirement when searching large datasets, especially across multiple data commons as it provides users with a structured filtering system to dynamically build cohorts based on user defined criteria. Beakerhub is unique as the interface enables scientists to remain actively involved in the process by reviewing results, providing feedback, and contributing their expertise. For example, upon user initiated query, the platform asks clarifying questions when needed to better understand user intent and improve the relevance and accuracy of outputs.

2.1 OBJECTIVE

The primary objective of this project is to develop, deploy, and maintain a Data Discovery Portal for the CRDC using Jataware's BeakerHub Platform through multiple phases. The Data Discovery Portal will provide users with powerful underlying AI technologies. The Data Discovery Portal will contain new AI and large language model functionality in conjunction with a traditional faceted search user interface. This comprehensive approach aims to create a user-friendly Data Discovery Portal that simplifies data access, enhances visualization, promotes collaboration, and democratizes understanding of cancer research data to enhance reusability and new hypothesis generation to further cancer research for new discoveries.

3.0 SCOPE – Base Period

In the non-severable two year base period, the Contractor shall develop a comprehensive CRDC Data Discovery Portal that integrates features from BeakerHub and traditional data exploration methods by leveraging rich data resources available in the CRDC ecosystem including the CDA search database that provides aggregated metadata from across the CRDC, and the CDA search API allowing users to query and build cohorts based on various search criteria (disease, study, gene IDs, variants, etc.). Based on the aggregation, transformation, and manual harmonization activities described above, CDA serves as a source of ground truth for CRDC. However, as described below under Data Sources, other integrations should be evaluated over the course of the contract period with the NCI federal lead.

Accessing, analyzing, or downloading controlled-access data is out of scope for the minimal viable product (MVP). However, the Data Discovery Portal shall allow users to hand off their cohort search results, including those from controlled studies, to the NCI Cloud Resource platform to grant access and enable analysis in a secure, FISMA Moderate system provided by the NCI Cloud Resource. Depending on future requirements set by NCI, the portal may implement authorization (AuthZ) via NIH RAS and DCF to expand the portal's ability to grant access to controlled data sets to users who have received approval from NIH dbGaP.

3.1 Project Description and Requirements

A phased approach will be used to develop and deploy the CRDC Data Discovery Portal, enabling incremental delivery and risk reduction. The work shall begin with the instantiation of BeakerHub, hosted by Jataware, which provides access to CRDC data sources to support initial design and validation activities. Users will be able to locate Jataware hosted BeakerHub on the current CRDC website for the initial pilot release. Future work shall include the development of a faceted search portal leveraging the Cancer Data Aggregator (CDA) API and transition of the BeakerHub platform to an NCI-hosted environment. In the final phase, the Contractor shall deliver a fully integrated, NCI-hosted production Data Discovery Portal combining the search functionality and full features of the BeakerHub platform, subject to stakeholder acceptance and applicable authorization requirements.

3.2 Data Sources

BeakerHub currently integrates over 20 data resources. For the MVP, the Contractor shall provide the capability to explore, analyze, and reason over data from all CRDC data commons, with CDA providing the primary source of truth for cross-CRDC metadata. Additional data sources, e.g., current or new CRDC integrations, should be evaluated for demonstrated added value to the end user. These additional data sources should already have a method to access data, such as an existing, documented API, as developing such access methods for these data sources is out of scope. In future phases, we anticipate integration of work currently being performed by additional ARPA-H performers as requirements are set by the NCI.

It should be noted that CDA performs transformation and harmonization on only a subset of data elements, representing the remaining data in its original format provided by the CRDC data commons. Manual data harmonization is out of scope for this project, and it is understood that some existing data conflicts may introduce difficulties for end users. However, it is likely that the current BeakerHub integration with ARPA-H BDF performer, Netrias', data harmonization model trained on CRDC may provide some additional level of automated harmonization. We anticipate that additional parallel efforts on solutions for data harmonization will feed into this work as integrations to be incorporated in future phases.

3.3 User Interface Key Features

An emphasis shall be placed on portal design and user experience so that it is easy to use and intuitive. Users must be able to run queries or make selections with minimal guidance. All portal functions will be included in documentation and training materials (e.g., user manuals, video tutorials, recorded webinars, and educational presentations) provided to support users.

3.4 Search Dashboard

A web-based interface will be built on the CDA API, providing faceted search across all CRDC Data Commons (GDC, PDC, IDC, etc) and other data sources deemed appropriate by the NCI. The dashboard shall include dynamic facets, query builders, dynamic data visualization, chat interface, cohort builder, and file export functionality to allow downstream analysis on NCI Cloud Resources. Selected data from the Data Discovery Portal will be exportable to the NCI Cloud Resource's Beakerhub Platform, providing users workflows for advanced analysis, preserving search context and filtering criteria. Search results will be transferred to the NCI Cloud Resource via the CRDC DRS style manifest. This manifest standard provides the NCI Cloud Resource with the required information necessary for the data to be imported securely into the cloud platform. This creates a continuous workflow from data discovery through sophisticated genomic and clinical analysis.

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To introduce AI-driven capabilities without increasing user burden, the Data Discovery Portal will align with design and navigation features of existing CRDC data portals, making advanced functionality accessible and easy to adopt. BeakerHub will support data discovery by providing users with a familiar, faceted search experience for exploring CRDC datasets. Its technology uniquely allows users to search using either natural language or faceted filters, automatically translating those inputs into Python code that runs within a Jupyter Notebook. The platform then interprets the results and presents them to users in plain language or through updates to the faceted display.

This approach enables users to perform searches and analyses with full transparency, as all generated code is completely accessible. Users can also connect their query results directly to BeakerHub's Jupyter Notebook environment, gaining access to citations, detailed methods, and resource attribution. By leveraging multiple components of the BeakerHub platform, CRDC search and analysis become savable, reproducible, and interoperable across the broader cancer research ecosystem. This functionality is essential for ensuring the accuracy and trustworthiness of AI-generated code and analyses in scientific discovery.

<u>Provided below are example usage for the Data Discovery Portal:</u>

Mixed-Initiative Interaction

Users will be able to seamlessly transition between conversational queries and GUI manipulation. For example:

- User asks: "Add lung cancer patients" (agent modifies filters)
- User manually adjusts age ranges in GUI (agent acknowledges change in conversation)
- User requests: "Why did my results change?" (agent explains filter modifications)

Progressive Discovery

Simple natural language queries should reveal increasingly sophisticated search options through the GUI, with the agent providing context-aware suggestions based on current search results. For example:

User is a cancer researcher with a hypothesis correlating a specific mutation with a specific kidney cancer morphology.

- User asks: Find subjects with kidney cancer having both imaging and sequenced mutation data
- Agent asks "Should I search broadly for "kidney cancer" or would you like me to focus on specific subtypes?" and updates the diagnosis type facets appropriately to include only those relevant facet options
- User selects facet(s) of interest in GUI, in this case kidney cancer
- Agent asks "Are you interested in any type of imaging (CT, MRI, etc.) or specific imaging modalities?" and updates the imaging modality and file type facets appropriately to include only those relevant facet options
- User selects facet(s) of interest in GUI, in this case "CT Image"
- Agent asks "Are you looking for any genomic sequencing data, specific types of mutations or specific genomic platforms or data types?" and

updates file type and file format facets appropriately to include only those relevant facet options
- User selects facet(s) of interest, in this case “somatic mutation”
- Agent provides an updated plan of action and asks if user would like to update any facets
- Agent begins the query, reporting out errors as it encounters them, providing user with the ability to view the code that the agent is running and providing natural language explanations for how it is resolving errors or modifying the plan of action.
- Agent provides a downloadable list of subjects with both CT files and mutation files
- User reviews output and asks agent to provide DRS ids for all files
- Agent provides a full drs listing in the SB-CGC required DRS manifest format, as well as outputting summary statistics, observation overview, assumptions and code references.

3.5 Analysis
While in-depth and computation intensive analytics are currently supported by the NCI Cloud Resource for open and controlled access data, lightweight analytics and data visualizations for open data shall be made available as part of the Data Discovery Portal. Data Discover Portal shall enable users to visualize summary metrics of user-selected cohorts with regard to the number of cases, samples, cancer, and data types. Where possible, analysis and visualization tools should be available directly from the Data Discovery Portal to the user, providing access to the NCI Cloud Resource only when additional computational resources and/or authorization (AuthN) is required.

3.6 Conversational AI Agent
The Contractor shall develop Data Discovery Portal conversational AI agent to be interactive with the traditional search dashboard. The conversational AI agent shall be developed leveraging context-aware language model with full visibility into the dashboard state and the ability to manipulate the interface programmatically. The agent will query current search filters, selected data, and session history while directly modifying interface elements based on natural language requests. The system will highlight anomalous patterns in data and provide natural language explanations of complex relationships across datasets and cohorts.

3.7 Interoperability and Authentication/Authorization
The CRDC ecosystem follows the FAIR data principles, making data findable, accessible, interoperable, and reusable. As part of CRDC, the Data Discovery Portal shall implement CRDC architecture standards to meet these principles. CRDC is a driver project of the Global Alliance for Genomics and Health (GA4GH), a worldwide consortium that publishes open technical and policy standards for responsible, interoperable sharing of genomics and health data. To align with GA4GH, the portal shall use published standards such as the Data Repository Service (DRS) for uniform data access and GA4GH Passports and Visas to manage user permissions. Authentication and

authorization (sign-in and access) will be provided through NIH's Researcher Auth Service (RAS), which offers secure single sign-on across NIH systems and implements the GA4GH Passport and Visa model. Adhering to GA4GH and RAS ensures secure, consistent access to NIH's open and controlled datasets, improves the user experience, and makes the portal interoperable with the broader CRDC ecosystem and NIH programs, such as the NIH Cloud Platform Interoperability (NCPI) initiative.

Additional examples of in-scope solutions to expand CRDC's data interoperability across the cancer research community are:

- MCP Server: The creation of a CRDC MCP server to enable other analysis platforms to leverage the portal's search capabilities and access selected datasets programmatically.

- Export Flexibility Multiple export formats (TSV, JSON, DataFrame serialization, etc) support integration with existing research workflows and tools.

3.8 Security

Post-MVP Data Discovery Portal must comply with FISMA Low security requirements initially, as the data to be queried and presented by Beakerhub Portal will only consist of open access data. No dbGaP controlled access data can be stored and/or redistributed through BeakerHub's storage system. The FISMA level for the Data Discovery Portal may be elevated to Moderate in the future, depending on portal requirements determined by the NCI.

4.0 SCOPE – Option Periods

During Option Period 1 and Option Period 2, the Contractor shall provide ongoing Operations and Maintenance (O&M), performance monitoring, corrective and preventive maintenance, enhancement support, documentation updates, security and compliance support, and user assistance for the Data Discovery Portal. The Contractor shall preform these services in a manner that supports system availability, reliability, security, scalability and continues improvement, and shall produce the associated reports, plans, documentation, and other deliverables identified in the Deliverables Table in section 6.2.

4.1 O&M Planning and Governance

The Contractor shall maintain and update the O&M Plan to reflect the current production environment, system architecture, operational procedures, maintenance activities, monitoring approach, incident response processes, security requirements, and support workflows. The O&M Plan shall define the Contractor's approach to day-to-day operations, issue resolution, patch management, performance monitoring, documentation maintenance, and user support during each Option Year.

Deliverable:

- Updated O&M Plan: Submitted within 30 days of the start of each Option Year and updated as needed

4.2 System Operations, Monitoring and Reporting

The Contractor shall provide day-to-day system operations support, including continuous monitoring of system health, uptime, availability, performance, logs, and operational metrics across all hosted environments. The Contractor shall identify, investigate, and resolve issues affecting system functionality, integrated search services, AI-supported capabilities, infrastructure performance, and user access.

The Contractor shall track and report on system operations, including uptime, incidents, response and resolution status, performance trends, maintenance activities, usage patterns, and identified risks. Reporting shall include routine monthly status updates and quarterly summaries of trends, performance, and improvement actions.

Deliverable:

- Monthly O&M and Performance Reports: Document system status, uptime, incidents, maintenance activities, and operational performance.
- Quarterly O&M and Performance Reports: Summarize overall system health, usage trends, risks, and improvement actions.

4.3 Corrective, Preventive and Adaptive Maintenance

The Contractor shall perform the corrective, preventive, and adaptive maintenance needed to keep the system stable and improve it over time. This includes identifying and resolving defects, bugs, failed processes, integration issues, and security vulnerabilities; applying patches, updates, and security fixes; and performing preventive maintenance to reduce downtime and improve long-term reliability.

The Contractor shall also support system enhancements and continuous improvement by maintaining a prioritized backlog of proposed fixes, changes, and enhancements based on user feedback, operational data, performance findings, and Government direction. The Contractor shall maintain a release roadmap that identifies planned releases, timelines, dependencies, and priorities.

Deliverables:

- Enhancement Backlog Updates: Maintained continuously and formally submitted monthly.
- Release Roadmap Updates: Submitted quarterly or as needed before major release cycles.
- Release Notes: Provided with each production deployment and shall summarize the fixes, updates, enhancements, system impacts, and user-facing changes included in the release.

4.4 Performance Optimization

The Contractor shall regularly assess and improve system performance. This includes monitoring and improving query execution, response times, reliability, scalability, and cloud resource utilization. The Contractor shall also assess and improve the performance of CDA-integrated search services, AI-supported features, and related infrastructure to reduce latency, improve accuracy, increase throughput, and enhance the user experience.

Performance findings and improvement actions shall be documented in routine O&M reports and used to inform backlog priorities, release planning, and annual recommendations.

Deliverables:
- Monthly and Quarterly O&M and Performance Reports
- Enhancement Backlog Updates
- Release Roadmap Updates
- Annual System Performance & Improvement Report

4.5 Security and Compliance Support

The Contractor shall support ongoing security and compliance activities needed to maintain the system in accordance with applicable NIH, NCI, and Federal requirements. This includes security monitoring, vulnerability identification and remediation, patch management, log review, audit support, and maintenance of the documentation needed to support FISMA Low or Moderate compliance, as applicable, and continued Authority to Operate (ATO) status.

The Contractor shall ensure that system operations and maintenance remain aligned with NIH Researcher Auth Service (RAS), GA4GH standards, CRDC security architecture, and other applicable Government security and interoperability requirements. Any security-related or architectural changes shall be reflected in the O&M Plan and related system documentation.

Deliverables:
- O&M Plan
- System Documentation Updates
- Monthly and Quarterly O&M and Performance Reports
- Release Notes: As applicable for security fixes and compliance-related change

4.6 System Documentation and User Documentation

The Contractor shall maintain accurate and up-to-date technical and operational documentation reflecting system changes, enhancements, interfaces, configurations, workflows, and maintenance activities. Documentation shall be updated after major releases, architectural changes, or other significant updates.

The Contractor shall also maintain end-user documentation and training materials, including user guides, FAQs, onboarding materials, release communications, and other resources needed to support effective use of the system. User-facing documentation shall reflect current functionality and comply with Section 508 accessibility requirements, as applicable.

Deliverables:

- System Documentation Updates: Provided on an ongoing basis and within 10 business days of major updates.
- User Documentation & Training Materials: Updated with each major feature release or quarterly, as applicable.

4.7 User Support and Training

The Contractor shall provide technical support for users and stakeholders, including the intake, tracking, and resolution of issues related to portal functionality, data access, workflows, search services, and AI-supported capabilities. The Contractor shall maintain support resources such as knowledge base content, FAQs, and other user support materials.

As needed and in coordination with the Government, the Contractor shall provide training sessions, demonstrations, webinars, or other knowledge transfer activities to support adoption of new features, workflows, and enhancements.

Deliverables:

- User Documentation & Training Materials
- Training Sessions / Webinars: Scheduled as needed in coordination with the Government.
- Monthly O&M and Performance Reports**:** Including relevant support issue and status summaries.

4.8. Annual Performance Review and Continuous Improvement

Near the end of each option year, the Contractor shall perform an annual review of system performance, operational maturity, enhancement progress, user adoption, service reliability, and opportunities for improvement. Based on that review, the Contractor shall provide recommendations for future enhancements, technical improvements, operational refinements, and support priorities for the next performance period.

Deliverables:

- Annual System Performance & Improvement Report: Submitted no later than 30 days before the end of the Option Year.

5.0 CONTRACT REQUIREMENTS/ AND PERSONNEL QUALIFICATIONS

STATEMENT OF WORK (SOW)

The Contractor shall perform the following tasks:

5.1 Project Management and Coordination

Project Management Plan

The Contractor shall develop and maintain a Project Management Plan (PMP) that describes the overall approach for managing and executing the requirements under this contract. The PMP shall include, at a minimum:

- Project goals and objectives
- Task descriptions and timelines
- Key milestones and deliverables
- Roles and responsibilities of contractor staff
- Communication plan, including frequency of meetings and reporting
- Risk management and mitigation strategies

The PMP shall be submitted to the Government for review and approval within 30 calendar days after award and updated as necessary throughout the period of performance.

5.2 Technical Support and Operations

The Contractor shall provide technical support services necessary to operate, maintain, and enhance the CRDC Data Discovery Portal. These services shall include, but are not limited to:

- System monitoring and routine maintenance
- Troubleshooting and resolution of technical issues
- Implementation of approved system enhancements or upgrades
- Ensuring system availability, performance, and reliability

All work shall be performed in accordance with applicable NIH/NCI IT security, privacy, and data management policies.

5.3 Data Management and Quality Assurance

The Contractor shall support data management activities related to the collection, validation, storage, and retrieval of data. This includes:

- Performing data quality checks and validation procedures
- Documenting data standards, definitions, and metadata
- Supporting data harmonization and integration activities, as applicable
- Ensuring data integrity and version control

The Contractor shall implement quality assurance procedures to ensure accuracy, completeness, and consistency of all data and deliverables.

5.4 Documentation and Reporting

The Contractor shall prepare and submit technical and progress documentation as required, including:

- Monthly and quarterly progress reports summarizing work performed, issues encountered, and planned activities
- Technical documentation describing system architecture, workflows, and changes implemented
- An annual report summarizing project performance, achievements, and an updated project roadmap identifying planned activities and priorities for the upcoming performance period.
- User guides or standard operating procedures (SOPs), if required

All documentation shall be submitted in electronic format and is subject to Government review and approval.

5.5 Meetings and Communication

The Contractor shall participate in meetings with Government staff, which may include:

- Kickoff meeting within two (2) weeks after award
- Regular monthly status meetings
- Ad hoc meetings as requested by the Government

Meeting summaries or minutes shall be provided to the Government within five (5) business days after each meeting.

5.6 Security, Privacy, and Compliance

The Contractor shall comply with all applicable Federal, HHS, NIH, and NCI policies and regulations, including but not limited to:

- Information security and privacy requirements
- Data use and confidentiality agreements
- Section 508 accessibility requirements (if applicable)

Contractor personnel shall complete all required security and privacy training prior to accessing Government systems or data.

5.7 Deliverables and Acceptance Criteria

STATEMENT OF WORK (SOW)

The Contractor shall deliver all products identified in Section 5.0 in accordance with the approved schedule. Deliverables shall be considered acceptable upon written approval by the Technical Point of Contact (TPOC). Revisions shall be made at no additional cost to the Government until deliverables meet acceptance criteria.

6.0 REPORT(S)/DELIVERABLES AND DELIVERY SCHEDULE

Complete deliverables schedule listed below.

6.1 Report Deliverables

All written deliverable products shall be submitted in draft format for review, comment and approval by the Technical Point of Contact (TPOC) (replaces COR). Final copies of approved drafts shall be delivered to the TPOC within five (5) business days after receipt of the Government's comments.

All written draft and final deliverable products shall be submitted in electronic copy for review and comment. Other quantities and formats may be submitted after prior approval from the TPOC. Electronic copies shall be submitted in the most recent version of the Word format, unless prior approval for another format has been obtained from the TPOC.

All deliverables shall be sent electronically (Microsoft Word or Excel, unless approved by the TPOC) per the following deliverable schedule:

6.2 Deliverables Table

Base Year Deliverables Table

Task Area	Deliverable	Deliverable Description / Format Requirements	Due Date
5.1	Project Management Plan	Project Management Plan in Word or PDF format outlining governance, schedule, risk management, and communication approach.	30 calendar days after award
5.2	Technical Design & Development Documentation Package	Technical design document including site mock-ups and development plan with initial timelines; submitted in Word or PDF format.	Within 75 calendar days of contract start or as agreed upon with the TPOC
5.1	Agile Management Deliverables	Sprint retrospective summaries and sprint planning/backlog grooming documentation in Word or PDF format.	At the end of each sprint and no less frequently than quarterly, as agreed upon with the TPOC

STATEMENT OF WORK (SOW)

5.4	Monthly Status Report	Monthly performance and progress report summarizing activities, accomplishments, issues, risks, and planned next steps.	By the 10th day of each month or as agreed upon with the TPOC
5.4	Quarterly Status Report	Quarterly performance report replacing the Monthly Report when due, summarizing cumulative progress, metrics, and risk management.	By the 10th day of each quarter or as agreed upon with the TPOC
5.4	Annual Report and Project Roadmap	Annual report summarizing project performance, achievements, and updated project roadmap in Word or PDF format.	Annually on the contract start date or as agreed upon with the TPOC
5.4	Final Report	Comprehensive final report summarizing project outcomes, deliverables, lessons learned, and recommendations.	Within 30 days of the end of the agreement or as agreed upon with the TPOC
3.1	CRDC BeakerHub MVP Instance	Operational CRDC BeakerHub MVP instance hosted by Jataware, including system architecture diagram and supporting technical documentation.	Year 1, Quarter 1
3.4	Search Portal MVP & Gap Analysis Package	Search functionality gap analysis and mapping document; pilot faceted search portal MVP (source code and deployed instance); conversational AI agent design specification.	Year 1, Quarter 2
3.1	CBIIT-Hosted BeakerHub Test Environment Package	Configured and operational CBIIT-hosted BeakerHub instance for internal testing, including deployment/configuration documentation and training materials.	Year 1, Quarter 3
3.1	BeakerHub MVP Pilot Release & Documentation Package	MVP pilot release deployed to stakeholders, including pilot evaluation plan, technical documentation (architecture, data flow, security overview), and end-user documentation.	Year 1, Quarter 4

3.8	Security Compliance & ATO Package	Initiate FISMA Low compliance and ATO activities for the CBIIT-hosted BeakerHub platform and faceted search portal. Includes FISMA Low security package (as applicable), System Security Plan (SSP), and supporting ATO documentation (policies, procedures, and control mappings).	Year 2, Quarter 1
3.1	Enhancement & Release Management Package	Collect stakeholder feedback and implement approved enhancements and feature updates. Includes stakeholder feedback summary report, prioritized enhancement backlog, updated portal and BeakerHub releases, and release notes.	Year 2, Quarter 2
3.1	Production Release & Documentation Package	Release the public production version of the Data Discovery Portal integrated with the CBIIT-hosted BeakerHub platform. Includes public production release, public-facing documentation and FAQs, and Operations and Maintenance (O&M) guide.	Year 2, Quarter 3
3.2	Integration Assessment & Roadmap Package	Evaluate integration of additional data sources and future enhancements. Includes data source integration assessment report, technical integration recommendations, and integration roadmap.	Year 2, Quarter 4

Option Period Deliverables Table

TASK AREA	DELIVERABLE	DELIVERABLE DESCRIPTION / FORMAT REQUIREMENTS	DUE DATE
4.1	O&M Plan (Updated)	Updated plan describing system operations, monitoring, maintenance, security, and support processes. Submitted in Word or PDF format.	Within 30 days of Option Year start and as updated

4.2	O&M and Performance Reports (Monthly)	Monthly report detailing system operations, uptime, incidents, performance metrics, and maintenance activities. Submitted in Word or PDF format.	Within 10 business days after month-end
4.2	O&M and Performance Reports (Quarterly)	Quarterly report summarizing performance trends, system health, usage analytics, and improvement actions. Submitted in Word or PDF format.	Within 15 business days after quarter-end
4.3	Enhancement Backlog Updates	Prioritized list of enhancements based on user feedback, analytics, and Government direction. Submitted in Word or Excel format.	Updated continuously; formal submission monthly
4.3	Release Roadmap Updates	Planned schedule of upcoming releases and enhancements. Submitted in Word, Excel, or PowerPoint format.	Quarterly or prior to major release cycles
4.3	Release Notes	Documentation of all deployed updates, fixes, and enhancements. Submitted in Word or PDF format.	At time of each deployment
4.5	System Documentation Updates	Updated technical documentation reflecting system changes and enhancements. Submitted in Word or PDF format.	Within 10 business days of major updates
4.6	User Documentation & Training Materials	Updated user guides, FAQs, onboarding materials, and training content. Submitted in Word, PDF, or multimedia formats (508 compliant).	With each major feature release or quarterly
4.7	Training Sessions / Webinars	Conducted training sessions or webinars for users, including presentation materials and recordings.	Scheduled in coordination with the Government
4.8	Annual System Performance & Improvement Report	Comprehensive annual report on system performance, enhancements, user adoption, and recommendations. Submitted in Word or PDF format.	Within 30 days prior to Option Year end

7.0 Key Personnel and Qualifications

7.1 Level of Effort

The following personnel are considered Key Personnel essential to the successful performance of this contract. Key Personnel shall be assigned at the labor category and minimum level of effort (LOE) specified below. The Contractor shall not remove, replace, or divert Key Personnel without the prior written approval of the Contracting Officer (CO).

Requests for substitution shall be submitted in writing and shall include the proposed individual's qualifications demonstrating equal or superior education, experience, and technical expertise, as compared to the originally proposed Key Personnel.

Key Personnel Position	Labor Category	Minimum Level of Effort (FTE)	Annual Hours
Project Manager	Project Manager II	1.0	1920
Subject Matter Expert (Lead)	Subject Matter Expert II	1.0	1920

7.2 Qualifications

7.2.1 Project Manager (PM) – Project Manager II

Minimum Qualifications:

Education:

- Bachelor's degree in Business Administration, Project Management, Engineering, Information Systems, Public Health, Life Sciences, or a related discipline from an accredited institution.

Experience:

- A minimum of eight (8) years of progressively responsible experience managing complex projects of similar scope and complexity.
- At least five (5) years of experience managing Federal Government contracts, preferably within HHS/NIH or other scientific, research, or healthcare environments.
- Demonstrated experience leading multidisciplinary teams composed of technical, scientific, and IT professionals.

7.2.2 Subject Matter Expert (SME) II – Lead

Minimum Qualifications:

Education:

- Master's degree in a relevant scientific, technical, engineering, data science, public health, biomedical, or related discipline from an accredited institution.
    - A Bachelor's degree with an additional four (4) years of specialized experience may be substituted for the Master's degree.

Experience:

STATEMENT OF WORK (SOW)

- A minimum of ten (10) years of specialized professional experience in the applicable subject matter area relevant to the scope of work.
- At least five (5) years of experience providing expert technical or scientific leadership on Federal Government or large-scale research programs.